# Oral solids: capsules: HMTc Standard

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CITE AS

Pendergrass K. "Oral solids: capsules: HMTc Standard". Heavy Metal Certified, 2026. https://heavymetalcertified.com/standards/-supplements-oral-solids-capsules. CC BY 4.0.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified standard for Oral solids: capsules (Dietary Supplements (Human)), stated as sold. Every limit is read across from the strictest sovereign infant-and-child-food maximum level for the analyte and applied to all supplements as a voluntary child-protective over-compliance; the laxer adult supplement floor (sovereign supplement MLs, USP <2232>, Prop 65) is disclosed and never selected.

## 1 | How every number on this standard is set

HMTc Method v2.0 — child-protective over-compliance: strictest sovereign infant-and-child-food ML by lookup, applied to all supplements; adult floor (supplement ML / USP <2232> / Prop 65 MADL) disclosed, never selected

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 10 | human-grade read-across (Australia / New Zealand) |
| Cadmium (Cd) | Tier 1 | 5 | human-grade read-across (European Union) |
| Total arsenic (tAs) | Tier 2 | 50 | human-grade read-across (India) |

| | | | |
|---|---|---|---|
| Inorganic arsenic (iAs) | Tier 1 | 10 | human-grade read-across (European Union) |
| Mercury, total (tHg) | Tier 1 | 4 | human-grade read-across (Israel) |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 100 | human-grade read-across (European Union) |
| Tin (Sn) | Tier 2 | 5,000 | human-grade read-across (India) |
| Aluminium (Al) | Tier 2 | 500 | human-grade read-across (Australia / New Zealand) |
| Total chromium (Cr) | Tier 2 | 1,000 | human-grade read-across (China) |
| Hexavalent chromium (Cr-VI) | Tier 2 | ≤ Cr | bounded by Cr (reflex speciation) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 10 | Australia / New Zealand — infant formula products |
| Cadmium (Cd) | 5 | European Union — infant formulae, follow-on formulae, food for special medical purposes intended for infants and young children, and young-child formulae placed on the market as liquid and manufactured from cow's milk proteins or cow's milk protein hydrolysates |
| Total arsenic (tAs) | 50 | India — Infant Milk substitute and Infant foods |

| | | |
|---|---|---|
| Inorganic arsenic (iAs) | 10 | European Union — infant formulae, follow-on formulae, food for special medical purposes intended for infants and young children, and young-child formulae placed on the market as liquid |
| Mercury, total (tHg) | 4 | Israel — Infant and toddler formulas, ready-to-eat |
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 100 | European Union — infant/young-child formula, liquid |
| Tin (Sn) | 5,000 | India — Infant Milk substitute and Infant foods |
| Aluminium (Al) | 500 | Australia / New Zealand — infant formula, follow-on formula, and special medical purpose product for infants other than preterm-infant products |
| Total chromium (Cr) | 1,000 | China — human-food analog |
| Hexavalent chromium (Cr-VI) | ≤ Cr | Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell. |

## 4 | Cite this standard

Pendergrass, K. "Oral solids: capsules." In Heavy Metal Tested & Certified (HMTc) Dietary Supplements (Human) standard, version 1.0 (2026-08-14). Institute of Contaminant Standards (ICS), 2026.

## References


[1] Australia / New Zealand — human-food maximum levels
Binds Pb, Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] China — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Cd, iAs, Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] Israel — human-food maximum levels
Binds tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[5] India — human-food maximum levels
Binds tAs, Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

## STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

## COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

## DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/supplements--oral-solids-capsules. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

## PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/supplements--oral-solids-capsules.